Circular utilization of sugarcane bagasse for water and nutrient retention in two-type of sugarcane cultivation soil by biochar and hydrochar addition☆

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Highlights

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Biochar (400 °C) and hydrochar (160–180 °C) enhance soil water retention.

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Optimal effects were at 14 (sand) and 21 (clay loam) days, useful in moisture and flow resistance.

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Biochar and hydrochar positively impact soil physicochemical nutrient retention.

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Biochar and hydrochar improved the germination index in harsh situations.

Abstract

In this study, we investigated the potential of bagasse-derived biochar and hydrochar to enhance soil conditions for sugarcane cultivation. The physical and chemical properties of soils and plants were analyzed; the results indicated significant benefits of biochar and hydrochar in soil improvement, with varying effects based on the production parameters and soil types. Biochar and hydrochar reduced soil density by up to 3.73% in sand and 6.04% in clay loam, enhanced water retention by 59.48%–73.4% (Available water capacity), and improved nutrient retention by 58% (Total S) and 87% (Total P) in both sand and clay loam soils. Although both materials had a minimal impact on germination sett root development, they positively influenced the germination index of primary shoots under drought conditions. Biochar notably boosted sugarcane germination under normal

conditions, outperforming hydrochar; however, hydrochar showed greater support under drought conditions. These findings highlight the potential of bagasse-derived biochar and hydrochar as sustainable soil amendments for sugarcane cultivation, enhancing crop productivity and soil health.

Introduction

Brazil is the world's largest producer of sugarcane, followed by India, China, and Thailand [1]. In the 2019–2020 period, Brazil produced approximately 642.7 million tons of sugarcane, covering an area of 8.44 million hectares [1]. In Thailand, the supply of sugarcane to sugar refineries was 106 million tons in 2015 [2]. Various factors affect sugarcane production, including mechanized harvesting and extensive cultivation in areas with poor soil quality [1], [3]. Drought is a widespread problem for crops globally, inevitably affecting sugarcane yields [4]. For example, Thailand faced severe drought conditions during the period 2015–2017, resulting in a significant reduction in sugarcane yield, reaching 11 % in 2016 and 12.4 % in 2017 compared to 2015 [2]. This decline in sugarcane yield has negatively affected the sugar industry and its economic supply chains. Droughts cause severe problems, including water deficit stress, particularly during the early and middle growth stages of sugarcane [4]. To address this problem, the key to improving the yield of sugarcane production is enhancing soil quality, encompassing factors such as water and nutrients [5].

In sugar factories, approximately 0.3 tons of sugarcane bagasse waste are generated per ton of raw sugarcane [6]. Sugarcane bagasse can be used as a combustion fuel in power plants [7], biodegradable materials [8], reinforcement materials [9], feedstock for compost [10] and char for soil amendment [11]. In recent years, research has been conducted on the use of biochar and hydrochar for soil amendment [12], [13]. Bento et al. evaluated the nutrient release process of two types of hydrochar (230 °C, 13 h from sugarcane bagasse and vinasse) mixed with three types of soil. The condition of 4 % (w/w) mixing of hydrochar resulted in the highest release of organic carbon (C) and macronutrients. If used optimally, it can also serve as a fertilizer substitute [14]. Farid et al. investigated co-composted biochar from rice straw and sugarcane bagasse to improve soil properties; the authors found that all amendments could improve soil organic C with a C/N ratio (carbon to nitrogen ratio) of 11–28.5 and significantly increase zucchini seed germination performance [15]. Alves et al. reported that the mixture of sewage sludge and sugarcane bagasse biochar (pyrolysis at 450 °C, 30 min) increased total C by 27.8 %, which can improve overall soil fertility; however, the sugar beet production from the biochar-amended soil was lower than that of the conventional fertilized soil [16]. Sugarcane bagasse-derived biochar from top-lit updraft gasifier at 550 °C for 1 h was used for saline-sodic soil reclamation in Brazil, reducing soil salinity and improving seed germination and plant growth [17]. It has been reported that biochar from 600 °C pyrolysis of sugarcane bagasse with chemical fertilizer could increase the grain yield of maize and groundnut by 40 % compared to only chemical fertilization [18]. Hariyono et al. used sugarcane trash after harvesting biochar and compost to improve sand soil for sugarcane cultivation; the authors found that physical properties such as available water content, aggregate stability, and chemical properties Carbon (C), Nitrogen (N), Phosphorus (P), and Potassium (K) could be improved but did not significantly enhance the growth and yield of sugarcane [19]. Based on a recent literature review, studies to apply the practice of using biochar or hydrochar produced from bagasse to improve cultivation in sugarcane soil still have different techniques while a comprehensive comparison of biochar and hydrochar as soil amendments in two-type of sugarcane cultivation soils in Thailand has not been investigated in detail before. The concept presented in this study aligns with the principles of a circular economy, aiming for zero waste in sugar factories by converting sugarcane bagasse into char to support sugarcane cultivation land in proximity with the factory and ultimately improve the yield. This idea benefits sugar industries by increasing the value of waste material as well as reducing transportation and disposal costs through local processing at sugar mills. Improved soil properties, reduced fertilizer use, and higher crop yields can make the approach economically viable and sustainable. Due to the problems of weather change, drought, and a lack of soil moisture, which affect plant growth, adjusting the structure of the soil will positively affect the water retention of the soil and the nutrient retention available. To prove that concept, a germination test of sugarcane buds was performed using commercial sugarcane buds and two-type of real sugarcane cultivation soils in Thailand.

This study conducted tests and evaluations on char production through carbonization and hydrothermal carbonization on a laboratory scale, exploring variations in temperature and holding time across two types of sugarcane cultivation soils in Thailand. Comprehensive material analysis was carried out on the biochar and hydrochar. A series of experiments investigated soil water retention performance, water holding capacity, flow resistance capability, and soil nutrient retention performance. Finally, sugarcane bud germination tests and analysis of brix values were performed to assess the potential of using biochar or hydrochar as soil amendments for sugarcane production.

Sugarcane bagasse

Sugarcane bagasse was obtained from Mitr Phol Phu Khieo Bio-Power (Mitr Phol Bio-Power Co., Ltd., Chaiyaphum, Thailand) and used as the raw material. The raw material was collected during the last stage of ethanol production. Sugarcane bagasse was dried by outdoor sun drying at 27.5–34.5 °C for various days. The results of the sugarcane bagasse characterization are presented in Table 1S.

Conventional carbonization

The laboratory-scale carbonizer was made of stainless steel (SUS316) with a diameter of 45 mm and a height of

Effect of biochar and hydrochar on the density of soils

Table 2S shows the bulk density of raw soils and biochar/hydrochar mixed soil. The original bulk density for sand was 1.69 g/cm^3. When biochar was mixed with sand, the density consistently decreased, regardless of the biochar production conditions. Additionally, the addition of hydrochar to sand led to a noticeably reduction in the density of the mixture. This decrease in bulk density in sand with hydrochar addition resulted from the low density of hydrochar. The core method (diameter 25 mm,

Conclusions

In this study, we investigated sugarcane bagasse-derived biochar and hydrochar to improve two types of soil. The following conclusions were drawn:

- 1.

The biochar and hydrochar experiments showed reduced soil density, especially in sand soil (approximately 3.73 %). The high surface area (19.3 %) and C content (80.4 %) of the biochar played significant roles.

- 2.

Biochar and hydrochar enhance soil water retention, with effectiveness influenced by production temperature and soil type. Both biochar at

CRediT authorship contribution statement

Kor Taweengern: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tanaporn Thapsamut:** Investigation. **Chanoknunt Khaobang:** Investigation. **Chinnathan Areeprasert:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Surachet Aramrak:** Validation, Supervision,

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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